

Integral Calculus and Transform Dualities

for the Fabius–Rvachev System

Weighted antiderivatives, inverse-function identities, entire Mellin moments,
fractional primitives, order statistics, and infinite-sinc transforms

Research report prepared for the ProveIt Fabius-function program

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Abstract

This report develops an integral calculus for the Fabius function F , its inverse $Q = F^{-1}$, and Rvachev’s compactly supported function up . The starting point is the probability model

$$Y = \sum_{k \geq 1} 2^{-k} U_k, \quad Z = 2Y - 1,$$

where the U_k are independent uniform random variables on $[0, 1]$, F is the distribution function of Y , and up is the density of Z . This model turns integration into a combination of quantile transport, moment calculus, and infinite-product transforms.

The principal results are: a two-function inverse-pair integration identity containing all weighted area formulas for F and Q ; a bivariate Laplace–Fourier duality; an entire complex moment function $\mathfrak{M}(s) = \mathbb{E}Y^s = \int_0^1 Q(v)^s dv$ with two rapidly convergent Newton-type series; exact series for negative and fractional powers; a stable recurrence for negative moments and a positive normalized limit C_* , giving

$$\mathfrak{M}(-r) \sim C_* \frac{2^{r(r+1)/2}}{\Gamma(r)};$$

Riemann–Liouville primitives of arbitrary order for powers of F and Q ; beta-weighted formulas, Bernoulli-polynomial integrals, order-statistic representations, Cauchy transforms, and Gini/autocorrelation integrals expressed through the infinite sinc product. Numerical experiments are supplied in a reproducible Python script.

A post-snapshot Lean audit now provides the generic continuous-linear naturality and conditional operator/scalar affine-recursion substrate for weighted uniform laws, together with the named geometric weights $(1 - q)q^n$, their probability law, self-similarity, reflection, carried-unit-interval results, and the half-base bridges. This does not formalize the report’s inverse-pair, fractional-power, complex-Mellin, density, transform, moment, or fixed-point uniqueness consequences.

Here “new” means absent from the recursively audited LaTeX corpus of the referenced ProveIt directory at the time of writing. It is not a claim of worldwide priority. Each statement is labelled in context as proved, computational, or conjectural.

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1 Corpus audit, scope, and novelty boundary

1.1 What was reviewed

The repository was traversed recursively through the documentation root, including the main monographs, glossary, Lean walkthrough, non-elementarity article, papers, canonical research-frontier volume, current frontier drafts, and frozen archive. The archive map was used to identify historical duplicates and their canonical successors; living source files and current drafts were checked directly at theorem and section level, with targeted full-text searches for integrals, inverse-function area formulas, Mellin transforms, fractional integrals, order statistics, Stieltjes transforms, and negative moments. The repository’s own normalized audit records sixty-seven distinct LaTeX paths at one baseline snapshot, while the current tree contains later additions as well.

The following table states the boundary used in this report.

Imported baseline	How it is used here
Probability interpretation of F , compact support and refinement equations for up, and the infinite sinc product	Normalization and the common probabilistic language for all integrals.
Generic weighted-uniform laws and support-free quantile transport	The formal source now contains arbitrary Banach-valued distribution splitting and reflection, exact interval mass for non-negative real weights, continuous-linear and scalar naturality, conditional operator/scalar affine recursions, the named geometric-uniform probability law and its carried- $[0, 1]$ specialization, a support-free atom-compatible quantile pushforward, its compact specialization, and the clamped Fabius inverse theorem.
Exact dyadic values, rational moment recurrences, Bernoulli and q -binomial structure	Inputs to the new Mellin/Newton continuation; these arithmetic results are not reproved in full.
Lambert- W endpoint and inverse asymptotics, periodic phase corrections	Used only for comparison and future asymptotic directions. The negative-moment theorem below is instead obtained from an exact moment recurrence.
Dyadic self-sampling quadrature, finite-prefix inverse acceleration, Rvachev–Appell and inverse-moment polynomial systems	Explicitly excluded from novelty claims; only their standard Appell consequence for repeated primitives outside the support is recalled.
Earlier integration-frontier material on elementary weighted inverse identities	Treated as precursor material. The present contribution is the full two-function calculus, complex-parameter continuation, fractional-power duality, and transform consequences.
Normalized integer-order Volterra calculus and signed-global Fabius primitives	Now formalized independently in <code>NormalizedVolterra.lean</code> and <code>FabiusAntiderivatives.lean</code> : arbitrary-base oriented kernels and their algebra in real normed targets; literal iteration, strict FTC, the order semigroup, and Taylor reconstruction in complete targets; polynomial weights and exact dyadic Fabius primitive ladders; and later all-scale affine covariance and base-point transport.

Formalization update: what has crossed the frontier

At repository snapshot 0770442945d65bac5530c4f214c6c52c4cb9fbdc, Lean proves the complete *integer-order* substrate used by this report, in greater generality than the Fabius-only statements: the normalized Volterra operator and its algebra take values in an arbitrary real normed space, are based at an arbitrary real point, and use oriented endpoints. For continuous inputs in a complete target, literal iteration equals the Cauchy kernel; strict FTC, additive composition of orders, and Taylor reconstruction are formal. Kernel raising and finite polynomial commutators retain the weaker normed-space generality. The post-mainline checkpoint e109088ed17186e7fcff8043cb3c1d9b5a131b8e additionally proves division-free affine covariance for every real scale and exact base-point transport without any endpoint ordering. For every $n, p \in \mathbb{N}$, the signed extension has exact global formulas for $J_0^n \mathcal{F}$ and $J_0^n(t^p \mathcal{F}(t))$; the bounded F has the same formulas for $x \leq 1$.

The integer Volterra results do *not* by themselves formalize inverse-pair substitutions, noninteger

or complex powers, fractional Riemann–Liouville operators, negative-moment continuation, order statistics, or transform identities. Support-free measure-level quantile transport, together with its compact specialization, is now formal separately, as recorded below; the report’s stronger incomplete inverse-pair formulas remain research-frontier mathematics.

Results not found in the audited corpus

The following package appears to be new relative to the repository corpus: the universal (A, B) inverse-pair identity in [theorem 3.1](#); the entire Mellin function and its Newton/central series in [theorems 4.2](#) and [4.4](#); the mixed reflected Mellin function; the negative-moment recurrence, periodic limiting profile, and constant C_* in [theorems 5.1](#) and [5.2](#); the fractional duality of [theorem 7.1](#); beta and resolvent transforms; and the sinc-product Gini formula in [theorem 10.2](#).

1.2 Conventions

All powers of positive real variables use the real logarithm: $x^s = \exp(s \log x)$. The totalized Fabius distribution function is

$$F(x) = 0 \quad (x \leq 0), \quad F(x) = 1 \quad (x \geq 1).$$

Its inverse $Q : (0, 1) \rightarrow (0, 1)$ is the ordinary increasing inverse; endpoint values are understood by continuity. Angular frequency is used for Fourier transforms: $\widehat{f}(t) = \int_{\mathbb{R}} e^{itx} f(x) dx$.

2 Probability model and transform baseline

2.1 The dyadic uniform sum

Let $U_1, U_2, \dots$ be independent random variables uniformly distributed on $[0, 1]$, and put

$$Y = \sum_{k=1}^{\infty} \frac{U_k}{2^k}, \quad Z = 2Y - 1 = \sum_{k=1}^{\infty} \frac{2U_k - 1}{2^k}. \quad (1)$$

Then $Y \in [0, 1]$, $Z \in [-1, 1]$, and the laws are symmetric:

$$Y \stackrel{d}{=} 1 - Y, \quad Z \stackrel{d}{=} -Z. \quad (2)$$

The Fabius function is the distribution function of Y ,

$$F(x) = \mathbb{P}(Y \leq x), \quad (3)$$

and up is the density of Z . Thus

$$F'(x) = 2 \operatorname{up}(2x - 1), \quad \operatorname{up}(x) = \begin{cases} F(x + 1), & -1 \leq x \leq 0, \\ F(1 - x) = 1 - F(x), & 0 \leq x \leq 1, \\ 0, & |x| > 1. \end{cases} \quad (4)$$

In particular $F(x) + F(1 - x) = 1$ on $[0, 1]$, and the quantile obeys

$$Q(1 - v) = 1 - Q(v). \quad (5)$$

The self-decomposition

$$Y \stackrel{d}{=} \frac{U + Y'}{2} \quad (6)$$

holds with $U \sim \operatorname{Unif}[0, 1]$ independent of a copy Y' of Y . It will generate both the positive- and negative-moment recurrences.

Formal arbitrary-weight law underlying the dyadic split

The current Lean source proves the distributional mechanism before specializing the weights. For every norm-summable sequence in a real Banach space with its Borel structure, the weighted uniform-coordinate pushforward is a probability law, splits as an independent uniform head plus the shifted-tail law, and is invariant under reflection through half the total vector weight. Nonnegative real weights give mass one to their natural interval and mass zero to its complement. The law definition itself remains total even without measurability. More generally, the weighted head–tail joint-law theorem accepts an arbitrary independently chosen summable tail-weight sequence v . The shifted tail $v_n = w_{n+1}$ appears only when that theorem is specialized to split one parent law μ_w .

The declarations for this probability, splitting, reflection, and carried-set layer are
`Fabius.ProbabilityRepresentation.uniformProduct_map_head_tail_comp,`
`Fabius.ProbabilityRepresentation.weightedUniformDistribution,`
`Fabius.ProbabilityRepresentation.isProbabilityMeasure_weightedUniformDistribution,`
`Fabius.ProbabilityRepresentation.uniformProduct_map_head_tailWeightedUniformSeries,`
`Fabius.ProbabilityRepresentation.weightedUniformDistribution_split,`
`Fabius.ProbabilityRepresentation.weightedUniformDistribution_reflection,`
`Fabius.ProbabilityRepresentation.weightedUniformDistribution_Icc,` and
`Fabius.ProbabilityRepresentation.weightedUniformDistribution_compl_Icc.`
The exact dyadic-to-generic law bridge is `Fabius.ProbabilityRepresentation.weightedSumDistribution_eq_weightedUniformDistribution.`

The module now also proves the generic measure-theoretic engine behind affine self-similarity. If measurable g and Φ satisfy $f(\omega) = \Phi(\omega_0, g(\tau\omega))$ almost everywhere, then

$$f_*\mathbf{P} = \Phi_*(\lambda_{[0,1]} \otimes g_*\mathbf{P}),$$

with no separate measurability assumption on f ; a pointwise version is a corollary. For a norm-summable sequence in a real Banach space and a continuous linear map L , $L(S_w) = S_{Lw}$, and the corresponding Borel laws satisfy $L_*\mu_w = \mu_{Lw}$; the target normed space need not be complete. Pointwise scalar naturality needs neither summability nor completeness, whereas its law-level version retains source completeness, Borel structure, and summability. If $w_{n+1} = A(w_n)$ for a continuous linear endomorphism A , then

$$A_*\mu_w = \mu_{w+}, \quad S_w(\omega) = \omega_0 w_0 + A(S_w(\tau\omega)), \quad \mu_w = ((u, z) \mapsto uw_0 + Az)_*(\lambda_{[0,1]} \otimes \mu_w).$$

The scalar case $A = cid$ is valid for arbitrary real c , subject only to the displayed norm-summability hypothesis on the weights at the series-splitting and law levels.

The eleven declarations in this naturality tranche are `Fabius.ProbabilityRepresentation.uniformProduct_map_of_ae_head_tail_recursion,`
`Fabius.ProbabilityRepresentation.uniformProduct_map_of_head_tail_recursion,`
`Fabius.ProbabilityRepresentation.weightedUniformSeries_map,`
`Fabius.ProbabilityRepresentation.weightedUniformSeries_smul,`
`Fabius.ProbabilityRepresentation.weightedUniformDistribution_map,`
`Fabius.ProbabilityRepresentation.weightedUniformDistribution_smul,`
`Fabius.ProbabilityRepresentation.weightedUniformDistribution_map_eq_shift,`
`Fabius.ProbabilityRepresentation.weightedUniformSeries_split_of_shift_eq_map,`
`Fabius.ProbabilityRepresentation.weightedUniformDistribution_split_of_shift_eq_map,`
`Fabius.ProbabilityRepresentation.weightedUniformSeries_geometric_split,` and
`Fabius.ProbabilityRepresentation.weightedUniformDistribution_geometric_`

split. The last two generic names express only a conditional scalar weight shift; the named specialization is a separate layer.

The seven compatibility declarations are `Fabius.ProbabilityRepresentation.weightedUniformSeries_smul_weights`, `Fabius.ProbabilityRepresentation.uniformProduct_map_head_tail_function`, `Fabius.ProbabilityRepresentation.weightedUniformDistribution_isProbabilityMeasure`, `Fabius.ProbabilityRepresentation.weightedUniformDistribution_smul_weights`, `Fabius.ProbabilityRepresentation.uniformProduct_map_head_tail_weightedUniformSeries`, `Fabius.ProbabilityRepresentation.weightedUniformDistribution_unitInterval`, and `Fabius.ProbabilityRepresentation.weightedUniformDistribution_compl_unitInterval`. The law-level scalar compatibility identity is oriented as $\mu_{cw} = (x \mapsto cx)_* \mu_w$. For every real q , the named weights are $w_n(q) = (1 - q)q^n$, with exact zero and successor formulas. Under $|q| < 1$ they are norm-summable with total mass one; their series is continuous and measurable, and the probability law satisfies the exact head–tail product identity, affine self-similarity, and reflection. Under $0 \leq q < 1$, the weights are nonnegative, the series lies in $[0, 1]$, and the law gives mass one to that interval and zero to its complement. The nineteen declarations are `Fabius.ProbabilityRepresentation.geometricUniformWeight`, `Fabius.ProbabilityRepresentation.geometricUniformWeight_zero`, `Fabius.ProbabilityRepresentation.geometricUniformWeight_succ`, `Fabius.ProbabilityRepresentation.hasSum_geometricUniformWeight`, `Fabius.ProbabilityRepresentation.summable_norm_geometricUniformWeight`, `Fabius.ProbabilityRepresentation.geometricUniformWeight_nonneg`, `Fabius.ProbabilityRepresentation.geometricUniformSeries`, `Fabius.ProbabilityRepresentation.continuous_geometricUniformSeries`, `Fabius.ProbabilityRepresentation.measurable_geometricUniformSeries`, `Fabius.ProbabilityRepresentation.geometricUniformSeries_split`, `Fabius.ProbabilityRepresentation.geometricUniformSeries_reflect`, `Fabius.ProbabilityRepresentation.geometricUniformSeries_mem_Icc`, `Fabius.ProbabilityRepresentation.geometricUniformDistribution`, `Fabius.ProbabilityRepresentation.geometricUniformDistribution_isProbabilityMeasure`, `Fabius.ProbabilityRepresentation.uniformProduct_map_head_tail_geometricUniformSeries`, `Fabius.ProbabilityRepresentation.geometricUniformDistribution_selfSimilar`, `Fabius.ProbabilityRepresentation.geometricUniformDistribution_reflection`, `Fabius.ProbabilityRepresentation.geometricUniformDistribution_Icc`, and `Fabius.ProbabilityRepresentation.geometricUniformDistribution_compl_Icc`. The exact half-base bridges are `Fabius.ProbabilityRepresentation.weightedCoordinateSum_eq_geometricUniformSeries_one_half` and `Fabius.ProbabilityRepresentation.weightedSumDistribution_eq_geometricUniformDistribution_one_half`.

Still outside the formal layer are equality of topological support with $[0, 1]$, atomlessness, the q -specific CDF and density identities, characteristic-function products, moments and cumulants, and uniqueness of the affine measure fixed point.

2.2 Moment-generating and Fourier products

Define

$$P(t) = \mathbb{E}e^{tY}, \quad \mathcal{U}(t) = \mathbb{E}e^{tZ}. \quad (7)$$

Independence in [equation \(1\)](#) gives the locally uniformly convergent products

$$P(t) = \prod_{k=1}^{\infty} \frac{e^{t/2^k} - 1}{t/2^k}, \quad (8)$$

$$\mathcal{U}(t) = \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k}, \quad (9)$$

$$P(t) = e^{t/2} \mathcal{U}(t/2). \quad (10)$$

The characteristic function of the up law is therefore

$$\phi(t) = \widehat{\text{up}}(t) = \prod_{k=1}^{\infty} \frac{\sin(t/2^k)}{t/2^k}. \quad (11)$$

The scaling equation

$$P(2t) = \frac{e^t - 1}{t} P(t) \quad (12)$$

is the exponential generating-function form of the moment recurrence below.

2.3 Rational moments, cumulants, and Bell polynomials

Write

$$m_n = \mathbb{E}Y^n = \int_0^1 Q(v)^n dv, \quad \mu_n = \mathbb{E}Z^n = \int_{-1}^1 x^n \text{up}(x) dx. \quad (13)$$

Taking the n -th moment in [equation \(6\)](#) gives

$$(2^n - 1)m_n = \sum_{k=0}^{n-1} \binom{n}{k} \frac{m_k}{n - k + 1}, \quad m_0 = 1. \quad (14)$$

Thus every m_n is rational. The first values are

$$1, \frac{1}{2}, \frac{5}{18}, \frac{1}{6}, \frac{143}{1350}, \frac{19}{270}, \frac{1153}{23814}, \frac{583}{17010}, \dots \quad (15)$$

The logarithm of [equation \(9\)](#) and $\log(\sinh z/z) = \sum_{r \geq 1} 2^{2r-1} B_{2r} z^{2r} / (r(2r)!)$ give the cumulants

$$\begin{aligned} \kappa_1(Y) &= \frac{1}{2}, \\ \kappa_{2r}(Y) &= \frac{B_{2r}}{2r(2^{2r} - 1)}, \quad \kappa_{2r+1}(Y) = 0 \quad (r \geq 1), \end{aligned} \quad (16)$$

$$\begin{aligned} \kappa_{2r}(Z) &= \frac{2^{2r} B_{2r}}{2r(2^{2r} - 1)}, \quad \kappa_{2r+1}(Z) = 0. \end{aligned} \quad (17)$$

Consequently m_n and μ_n are complete exponential Bell polynomials in these cumulants. A useful bridge between the one-sided and centered moments is

$$\mu_{2j} = 2 \int_0^1 x^{2j} \text{up}(x) dx = \frac{2m_{2j+1}}{2j+1}, \quad \mu_{2j+1} = 0. \quad (18)$$

For example

$$\mu_2 = \frac{1}{9}, \quad \mu_4 = \frac{19}{675}, \quad \mu_6 = \frac{583}{59535}. \quad (19)$$

3 A universal integration calculus for F and Q

The classical area identity for a function and its inverse is only the first member of a much larger family. The following theorem is the main algebraic engine of the report.

Exact boundary of the formal quantile theorems

The foundational measure-transport theorem is support-free. Let μ be any probability measure on $\mathbb{R}$, and let Q be almost-everywhere measurable for uniform measure on $[0, 1]$. If, for every $x \in \mathbb{R}$,

$$Q(y) \leq x \iff y \leq \text{cdf}_\mu(x) \quad \text{for almost every } y \in [0, 1],$$

then $Q_*\lambda_{[0,1]} = \mu$. There is no support assumption on μ and no requirement that Q take its source into $[0, 1]$. A compact specialization replaces the almost-everywhere sublevel hypothesis by a pointwise order characterization on the unit square, together with a.e. support and interval preservation. Both theorems permit atoms.

The Fabius specialization uses the globally clamped inverse but restricts the uniform source to $[0, 1]$; every strongly measurable Banach-valued statistic has the corresponding integral identity, with no integrability assumption because both Bochner integrals are totalized.

The exact declarations are `Fabius.map_quantile_eq`, `Fabius.map_inverseCDF_volume_restrict_Icc`, `Fabius.map_fabiusInv_restrict_Icc_eq_weightedSumDistribution`, and `Fabius.integral_comp_fabiusInv_restrict_Icc_eq_weightedSumDistribution` in `QuantileTransport.lean`. The two-function identity below is still a frontier claim: it contains incomplete integration-by-parts information not implied by the pushforward equality alone.

Theorem 3.1 (two-function inverse-pair identity). *Let A, B be absolutely continuous on $[0, 1]$. For every $z \in [0, 1]$,*

$$A(z)B(F(z)) - A(0)B(0) = \int_0^z A'(x)B(F(x)) \, dx + \int_0^{F(z)} A(Q(v))B'(v) \, dv. \quad (20)$$

At $z = 1$,

$$A(1)B(1) - A(0)B(0) = \int_0^1 A'(x)B(F(x)) \, dx + \int_0^1 A(Q(v))B'(v) \, dv. \quad (21)$$

The formulas remain valid under the usual improper-integral hypotheses whenever an endpoint is singular.

Proof. Differentiate the product $A(x)B(F(x))$, integrate from 0 to z , and in the term containing $F'(x)$ use the increasing substitution $v = F(x)$, so that $x = Q(v)$. $\square$

Remark 3.2. No special functional equation of the Fabius function is needed in [theorem 3.1](#); it is a quantile identity. What is special here is that the moments and transforms of this particular quantile can be calculated by dyadic products and rational recurrences.

3.1 Power duality and logarithmic derivatives

Taking $A(x) = x^p$, $B(v) = v^q$ gives the following Mellin-type duality.

Corollary 3.3 (power duality). *Initially for $\Re p, \Re q > 0$,*

$$p \int_0^z x^{p-1} F(x)^q \, dx + q \int_0^{F(z)} Q(v)^p v^{q-1} \, dv = z^p F(z)^q. \quad (22)$$

In particular,

$$p \int_0^1 x^{p-1} F(x)^q \, dx + q \int_0^1 v^{q-1} Q(v)^p \, dv = 1. \quad (23)$$

Both sides admit meromorphic or entire continuation in parameters whenever the endpoint flatness cancels the apparent singularities.

For all logarithmic derivatives at once, define

$$J_{r,s}(p, q) = \int_0^1 x^{p-1} (\log x)^r F(x)^q (\log F(x))^s dx, \quad (24)$$

$$K_{r,s}(p, q) = \int_0^1 v^{q-1} (\log v)^s Q(v)^p (\log Q(v))^r dv. \quad (25)$$

Differentiating [equation \(23\)](#) gives, for $r + s > 0$,

$$pJ_{r,s} + rJ_{r-1,s} + qK_{r,s} + sK_{r,s-1} = 0, \quad (26)$$

with $pJ_{0,0} + qK_{0,0} = 1$. Equation [equation \(26\)](#) is a compact general formula for every mixed derivative of the two Mellin kernels.

3.2 A bivariate Laplace–Fourier identity

The exponential choice $A(x) = e^{sx}$, $B(v) = e^{tv}$ produces an especially simple transform relation.

Theorem 3.4 (bivariate inverse-pair transform). *For all $s, t \in \mathbb{C}$ and $z \in [0, 1]$,*

$$s \int_0^z e^{sx+tF(x)} dx + t \int_0^{F(z)} e^{tv+sQ(v)} dv = e^{sz+tF(z)} - 1. \quad (27)$$

Hence

$$s \int_0^1 e^{sx+tF(x)} dx + t \int_0^1 e^{tv+sQ(v)} dv = e^{s+t} - 1. \quad (28)$$

The identities are entire in (s, t) ; the cases $s = 0$ or $t = 0$ are understood by continuity.

Every mixed power integral involving x and $F(x)$, together with its quantile dual, is obtained by coefficient extraction from [equation \(28\)](#). Purely imaginary s, t give a bivariate Fourier identity.

3.3 Weighted antiderivatives

Taking $B(v) = v$ and a primitive A of a prescribed weight yields the useful master formula

$$\int_0^x A'(t) F(t) dt = A(x) F(x) - A(0) F(0) - \int_0^{F(x)} A(Q(v)) dv. \quad (29)$$

For monomial weights this becomes the promised indefinite integral.

Corollary 3.5 (monomial–Fabius antiderivative). *For $p \neq -1$, initially where the integrals converge and then by continuation,*

$$\int_0^x t^p F(t) dt = \frac{x^{p+1} F(x) - \int_0^{F(x)} Q(v)^{p+1} dv}{p+1}. \quad (30)$$

The value at $p = -1$ is removable:

$$\int_0^x \frac{F(t)}{t} dt = F(x) \log x - \int_0^{F(x)} \log Q(v) dv. \quad (31)$$

On $[0, 1]$, where $\text{up} = 1 - F$,

$$\int_0^x t^p \text{up}(t) dt = \frac{x^{p+1}(1 - F(x)) + \int_0^{F(x)} Q(v)^{p+1} dv}{p+1}. \quad (32)$$

More generally, from [equation \(22\)](#),

$$\int_0^x t^{p-1} F(t)^q dt = \frac{x^p F(x)^q}{p} - \frac{q}{p} \int_0^{F(x)} Q(v)^p v^{q-1} dv. \quad (33)$$

Thus products of F and the positive half of up reduce to beta-weighted quantile integrals after expanding $(1 - F)^n$.

4 The entire complex moment function

Definition 4.1. For $s \in \mathbb{C}$, set

$$\mathfrak{M}(s) = \mathbb{E}Y^s = \int_0^1 Q(v)^s dv. \quad (34)$$

For nonnegative integers, $\mathfrak{M}(n) = m_n$.

4.1 Entire continuation

The endpoint flatness of F is much stronger than what is needed for ordinary positive moments: for every $N > 0$,

$$F(x) = O_N(x^N) \quad (x \downarrow 0). \quad (35)$$

Theorem 4.2 (entire quantile Mellin function). *The function $\mathfrak{M}$ in [equation \(34\)](#) is entire. For every $r \geq 0$,*

$$\mathfrak{M}^{(r)}(s) = \int_0^1 Q(v)^s (\log Q(v))^r dv = \mathbb{E}[Y^s (\log Y)^r]. \quad (36)$$

In particular all negative moments and all logarithmic moments exist.

Proof. Let $K \subset \mathbb{C}$ be compact and choose $A > 0$ with $\Re s \geq -A$ on K . For every fixed r , the integrand is bounded in modulus by $Y^{-A} |\log Y|^r$. Integration by parts gives

$$\mathbb{E}Y^{-A} = 1 + A \int_0^1 x^{-A-1} F(x) dx < \infty,$$

using [equation \(35\)](#) with an exponent larger than A ; inserting powers of $|\log x|$ does not alter convergence. Dominated differentiation on compact sets proves holomorphy and [equation \(36\)](#). $\square$

A first transform triangle follows immediately:

$$\int_0^1 x^{s-1} F(x) dx = \frac{1 - \mathfrak{M}(s)}{s}, \quad (37)$$

$$\int_0^1 x^{s-1} \text{up}(x) dx = \frac{\mathfrak{M}(s)}{s}, \quad (38)$$

$$\int_0^1 Q(v)^{s-1} dv = \mathfrak{M}(s-1). \quad (39)$$

The first quotient has a removable singularity at $s = 0$ and is entire; the second is the meromorphic continuation of the classical Mellin transform of $\text{up}|_{[0,1]}$ and has exactly one pole, at $s = 0$, with residue 1. Evenness gives

$$\int_{-1}^1 |x|^{s-1} \text{up}(x) dx = \frac{2\mathfrak{M}(s)}{s}. \quad (40)$$

4.2 Superpolynomial moment decay

Lemma 4.3. *For every $N > 0$,*

$$m_k = O_N(k^{-N}) \quad (k \rightarrow \infty). \quad (41)$$

Proof. For $k > 0$, the tail formula and symmetry give

$$m_k = k \int_0^1 x^{k-1} (1 - F(x)) \, dx = k \int_0^1 x^{k-1} F(1 - x) \, dx.$$

By [equation \(35\)](#), $F(1 - x) \leq C_N(1 - x)^N$, hence

$$m_k \leq C_N k B(k, N + 1) = O(k^{-N}).$$

□

4.3 Two globally convergent Newton series

Theorem 4.4 (Newton and central-even series). *Both of the following series converge locally uniformly for all $s \in \mathbb{C}$:*

$$\mathfrak{M}(s) = \sum_{k=0}^{\infty} (-1)^k \binom{s}{k} m_k, \quad (42)$$

$$\mathfrak{M}(s) = 2^{-s} \sum_{j=0}^{\infty} \binom{s}{2j} \mu_{2j} = 2^{-s} \sum_{j=0}^{\infty} \binom{s}{2j} \frac{2m_{2j+1}}{2j+1}. \quad (43)$$

They therefore represent the entire function $\mathfrak{M}$ on the whole plane.

Proof. For $0 < Y < 1$, expand $Y^s = (1 - (1 - Y))^s$. Taking expectations and using $1 - Y \stackrel{d}{=} Y$ gives [equation \(42\)](#). On every compact set in s , the binomial coefficients have at most polynomial growth in k ; [lemma 4.3](#) therefore justifies absolute, locally uniform convergence and interchange of sum and expectation.

For [equation \(43\)](#), write $Y = (1 + Z)/2$, expand $(1 + Z)^s$, and use the evenness of the law of Z . The relation [equation \(18\)](#) and [lemma 4.3](#) again give locally uniform convergence. □

Several useful specializations are immediate. For an integer $r \geq 1$,

$$\mathfrak{M}(-r) = \sum_{k=0}^{\infty} \binom{r+k-1}{k} m_k. \quad (44)$$

Every term is rational and positive. In particular,

$$\mathfrak{M}(-1) = \sum_{k=0}^{\infty} m_k. \quad (45)$$

Differentiation of [equation \(42\)](#) at the origin yields logarithmic moments in terms of signed Stirling numbers of the first kind $s(k, r)$:

$$\mathfrak{M}^{(r)}(0) = r! \sum_{k=r}^{\infty} \frac{(-1)^k s(k, r)}{k!} m_k. \quad (46)$$

For $r = 1$,

$$-\mathbb{E} \log Y = \sum_{k=1}^{\infty} \frac{m_k}{k} = \int_0^1 \frac{F(x)}{x} \, dx. \quad (47)$$

Numerically,

$$-\mathbb{E} \log Y = 0.758240000404408656 \dots \quad (48)$$

4.4 A mixed reflected Mellin function

The inverse symmetry [equation \(5\)](#) suggests the two-variable transform

$$\mathfrak{M}_2(a, b) = \int_0^1 Q(v)^a Q(1-v)^b dv = \mathbb{E}[Y^a(1-Y)^b]. \quad (49)$$

Proposition 4.5. *The function $\mathfrak{M}_2$ is entire on $\mathbb{C}^2$, symmetric in a, b , and has the locally uniformly convergent expansion*

$$\mathfrak{M}_2(a, b) = \sum_{k=0}^{\infty} (-1)^k \binom{b}{k} \mathfrak{M}(a+k). \quad (50)$$

It also has the central expansion

$$\mathfrak{M}_2(a, b) = 2^{-a-b} \sum_{j,k \geq 0} (-1)^k \binom{a}{j} \binom{b}{k} \mu_{j+k}. \quad (51)$$

If either parameter is a nonnegative integer, the corresponding binomial expansion terminates.

Proof. Endpoint flatness at both 0 and 1 gives local domination for arbitrary negative real parts, proving entire dependence on two parameters. Expand $(1-Y)^b$ to obtain [equation \(50\)](#); superpolynomial decay of shifted moments gives local uniformity. The central formula follows from $Y = (1+Z)/2$, $1-Y = (1-Z)/2$. $\square$

For example,

$$\int_0^1 Q(v)Q(1-v) dv = \frac{2}{9}, \quad \int_0^1 Q(v)^2 Q(1-v)^2 dv = \frac{34}{675}, \quad (52)$$

and

$$\int_0^1 \frac{dv}{Q(v)(1-Q(v))} = 2\mathfrak{M}(-1). \quad (53)$$

5 Negative moments: an exact recurrence and a new asymptotic constant

The Newton series [equation \(44\)](#) is effective for moderate negative orders. A second construction is much more stable for large orders and reveals a new normalization law.

5.1 The recurrence

For $s > 0$, define

$$A(s) = \mathfrak{M}(-s) = \mathbb{E}Y^{-s}, \quad D(s) = \mathbb{E}(1+Y)^{-s}. \quad (54)$$

Conditioning in [equation \(6\)](#) and integrating the uniform variable gives the following exact relation.

Theorem 5.1 (negative-moment recurrence). *For every real $s > 0$,*

$$A(s+1) = \frac{2^{s+1}}{s} (A(s) - D(s)). \quad (55)$$

Equivalently, for $s > 1$,

$$A(s) = \frac{2^s}{s-1} (A(s-1) - D(s-1)). \quad (56)$$

Moreover $D(s)$ has the rapidly convergent centered series

$$D(s) = \left(\frac{2}{3}\right)^s \sum_{j=0}^{\infty} \frac{(s)_{2j}}{(2j)! 3^{2j}} \mu_{2j}, \quad (57)$$

where $(s)_n = s(s+1) \cdots (s+n-1)$.

Proof. By [equation \(6\)](#), for $s > 0$,

$$A(s+1) = 2^{s+1} \mathbb{E} \int_0^1 (U + Y')^{-s-1} dU.$$

For $y > 0$,

$$\int_0^1 (u+y)^{-s-1} du = \frac{y^{-s} - (1+y)^{-s}}{s},$$

which gives [equation \(55\)](#). For [equation \(57\)](#), write $1+Y = (3+Z)/2$, expand $(1+Z/3)^{-s}$, and use the vanishing of odd moments. The factor 3^{-2j} makes the series geometrically stable in addition to the superpolynomial decay of μ_{2j} . $\square$

The first values generated by this recurrence are

$$\begin{aligned} A(1) &= 2.31488614400581905 \dots, \\ A(2) &= 6.55899183845822858 \dots, \\ A(3) &= 24.3890791927221220 \dots, \\ A(4) &= 128.368907270620532 \dots, \\ A(5) &= 1025.15473735960333 \dots \end{aligned} \quad (58)$$

5.2 A normalized limit and a periodic phase profile

Define

$$H(s) = \frac{\Gamma(s)A(s)}{2^{s(s+1)/2}}, \quad \rho(s) = \frac{D(s)}{A(s)}. \quad (59)$$

Then [equation \(55\)](#) becomes

$$H(s+1) = H(s)(1 - \rho(s)). \quad (60)$$

Both factors are positive. Since $\mathbb{P}(Y \leq 1/2) = 1/2$,

$$A(s) \geq 2^{s-1}, \quad D(s) \leq 1, \quad 0 < \rho(s) \leq 2^{1-s}. \quad (61)$$

This elementary estimate is enough to prove convergence.

Theorem 5.2 (periodic limiting profile). *For $0 < \theta \leq 1$, the product*

$$C(\theta) = H(\theta) \prod_{j=0}^{\infty} (1 - \rho(\theta + j)) \quad (62)$$

converges to a positive number. It extends by $C(\theta+1) = C(\theta)$ to a positive, continuous, one-periodic function. As $n \rightarrow \infty$,

$$A(n+\theta) = C(\theta) \frac{2^{(n+\theta)(n+\theta+1)/2}}{\Gamma(n+\theta)} (1 + O_{\theta}(2^{-n})). \quad (63)$$

For integer $r \rightarrow \infty$,

$$\boxed{\mathfrak{M}(-r) = C_* \frac{2^{r(r+1)/2}}{\Gamma(r)} (1 + O(2^{-r}))}, \quad C_* = C(1) > 0. \quad (64)$$

Proof. Iteration of [equation \(60\)](#) gives

$$H(n + \theta) = H(\theta) \prod_{j=0}^{n-1} (1 - \rho(\theta + j)).$$

The sum of the tail bounds in [equation \(61\)](#) is geometric. Hence the infinite product converges to a positive limit, and its omitted tail differs from 1 by $O_\theta(2^{-n})$. This proves [equation \(63\)](#). Shifting the initial point by one and using [equation \(60\)](#) proves exact periodicity. Uniform convergence on compact subintervals gives continuity; exact periodicity then removes the apparent endpoint at $\theta = 0$. The integer formula is the case $\theta = 1$. $\square$

For integers the normalized sequence

$$C_r = \frac{(r-1)! \mathfrak{M}(-r)}{2^{r(r+1)/2}} \quad (65)$$

is strictly decreasing and satisfies

$$C_{r+1} = C_r \left(1 - \frac{D(r)}{A(r)}\right), \quad C_r \downarrow C_*. \quad (66)$$

The product representation is

$$C_* = \frac{\mathfrak{M}(-1)}{2} \prod_{j=1}^{\infty} \left(1 - \frac{D(j)}{A(j)}\right). \quad (67)$$

The computation described in [section 12](#) gives

$$C_* = 0.750721980407870959 \dots \quad (68)$$

The rapid stabilization is visible in

r	1	2	4	6	8	12
C_r	1.15744307	0.81987398	0.75216157	0.75072879	0.75072199	0.7507219804

5.3 Logarithmic asymptotics and entire-function growth

Stirling's expansion in [equation \(64\)](#) gives a complete algebraic asymptotic series:

$$\begin{aligned} \log \mathfrak{M}(-r) &= \frac{\log 2}{2} r^2 - r \log r + \left(1 + \frac{\log 2}{2}\right) r + \frac{1}{2} \log r + \log \frac{C_*}{\sqrt{2\pi}} \\ &\quad - \frac{1}{12r} + \frac{1}{360r^3} - \frac{1}{1260r^5} + \dots, \end{aligned} \quad (69)$$

where

$$\log \frac{C_*}{\sqrt{2\pi}} = -1.20565842815389633 \dots \quad (70)$$

This refines the growth one would obtain by a direct saddle-point transfer from the Lambert- W endpoint expansion.

There is also a clean entire-function consequence. If $M_{\mathfrak{M}}(R) = \max_{|s|=R} |\mathfrak{M}(s)|$, then

$$|\mathfrak{M}(s)| \leq \mathbb{E} Y^{\Re s} \leq \mathbb{E} Y^{-R} = \mathfrak{M}(-R), \quad |s| = R, \quad (71)$$

while equality occurs at $s = -R$. Thus

$$M_{\mathfrak{M}}(R) = \mathfrak{M}(-R). \quad (72)$$

Corollary 5.3. *The entire function $\mathfrak{M}$ has order 2 and type $(\log 2)/2$.*

5.4 Singular Fabius integrals

Putting $s = p + 1$ in [equation \(37\)](#) shows that

$$\int_0^1 x^p F(x) dx = \frac{1 - \mathfrak{M}(p + 1)}{p + 1} \quad (73)$$

is an entire function of p , despite the apparent pole at $p = -1$. Therefore all negative powers are integrable against F . For integers $k \geq 2$,

$$\int_0^1 \frac{F(x)}{x^k} dx = \frac{\mathfrak{M}(1 - k) - 1}{k - 1}. \quad (74)$$

The first examples are

$$\begin{aligned} \int_0^1 \frac{F(x)}{x} dx &= 0.758240000404408656 \dots, \\ \int_0^1 \frac{F(x)}{x^2} dx &= 1.31488614400581905 \dots, \\ \int_0^1 \frac{F(x)}{x^3} dx &= 2.77949591922911429 \dots, \\ \int_0^1 \frac{F(x)}{x^4} dx &= 7.79635973090737400 \dots \end{aligned} \quad (75)$$

By [equation \(64\)](#),

$$\int_0^1 \frac{F(x)}{x^k} dx \sim C_* \frac{2^{k(k-1)/2}}{\Gamma(k)}. \quad (76)$$

This is an asymptotic law for a natural family of increasingly singular antiderivatives.

6 Definite weighted integrals

6.1 Polynomial and analytic weights

Let g be integrable and let $G(x) = \int_0^x g(t) dt$. Taking $A = G$, $B(v) = v$ in [theorem 3.1](#) gives

$$\int_0^1 g(x) F(x) dx = G(1) - \mathbb{E}G(Y), \quad (77)$$

$$\int_0^1 g(x)(1 - F(x)) dx = \mathbb{E}G(Y). \quad (78)$$

For a polynomial g , every quantity on the right is a rational combination of the moments m_n . For an entire g , the same formulas give convergent series whenever the Taylor expansion can be integrated termwise.

The monomial cases are

$$\int_0^1 x^p F(x) dx = \frac{1 - \mathfrak{M}(p + 1)}{p + 1}, \quad (79)$$

$$\int_0^1 x^p \text{up}(x) dx = \frac{\mathfrak{M}(p + 1)}{p + 1} \quad (\text{up} = 1 - F \text{ on } [0, 1]). \quad (80)$$

For $p \in \mathbb{N}$ these are rational. For example,

p	0	1	2	3
$\int_0^1 x^p F(x) \, dx$	1/2	13/36	5/18	1207/5400
$\int_0^1 x^p \text{up}(x) \, dx$	1/2	5/36	1/18	143/5400

For symmetric weights $g(x) = g(1 - x)$, symmetry alone gives

$$\int_0^1 g(x) F(x) \, dx = \frac{1}{2} \int_0^1 g(x) \, dx. \quad (81)$$

6.2 Bernoulli-polynomial weights

Let

$$\beta_n = \mathbb{E} B_n(Y). \quad (82)$$

Their exponential generating function follows from [equation \(8\)](#):

$$\sum_{n=0}^{\infty} \beta_n \frac{t^n}{n!} = \frac{tP(t)}{e^t - 1}. \quad (83)$$

Symmetry implies $\beta_{2j+1} = 0$. Since $B'_{n+1} = (n+1)B_n$, [equation \(77\)](#) gives, for $n \geq 1$,

$$\int_0^1 B_n(x) F(x) \, dx = -\frac{\beta_{n+1} - B_{n+1}}{n+1}. \quad (84)$$

The first nontrivial expected Bernoulli values are

$$\beta_2 = -\frac{1}{18}, \quad \beta_4 = \frac{23}{1350}, \quad \beta_6 = -\frac{1543}{119070}, \quad \beta_8 = \frac{1196113}{65063250}. \quad (85)$$

This supplies a direct integral bridge between the Bernoulli-polynomial structure and the Fabius distribution.

6.3 Beta weights

For $\Re a, \Re b > 0$, define

$$I_F(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} F(x) \, dx. \quad (86)$$

Theorem 6.1 (beta-weighted Fabius integrals). *For $\Re a, \Re b > 0$,*

$$I_F(a, b) = \mathbb{E} B_Y(b, a) = \sum_{n=0}^{\infty} \frac{(1-a)_n}{n!(b+n)} \mathfrak{M}(b+n), \quad (87)$$

$$\int_0^1 x^{a-1} (1-x)^{b-1} (1-F(x)) \, dx = \mathbb{E} B_Y(a, b) = \sum_{n=0}^{\infty} \frac{(1-b)_n}{n!(a+n)} \mathfrak{M}(a+n). \quad (88)$$

Here B_y is the incomplete beta function. If a in [equation \(87\)](#), or b in [equation \(88\)](#), is a positive integer, the corresponding series terminates. If both parameters are positive integers, the integrals are rational.

Proof. Use $F(x) = \mathbb{E} \mathbf{1}_{\{Y \leq x\}}$ and Fubini:

$$I_F(a, b) = \mathbb{E} \int_Y^1 x^{a-1} (1-x)^{b-1} \, dx = \mathbb{E} B_{1-Y}(b, a) = \mathbb{E} B_Y(b, a),$$

where the last equality uses $1 - Y \stackrel{d}{=} Y$. The hypergeometric expansion

$$B_y(b, a) = \sum_{n=0}^{\infty} \frac{(1-a)_n}{n!(b+n)} y^{b+n}$$

gives [equation \(87\)](#). The endpoint flatness and [lemma 4.3](#) justify convergence beyond the ordinary boundary restrictions of the pointwise hypergeometric series. The second identity is analogous. $\square$

For example, symmetry or direct termination gives

$$\int_0^1 x(1-x)F(x) \, dx = \frac{1}{12}. \quad (89)$$

Differentiating [equation \(87\)](#) in a and b produces all beta-logarithmic integrals with powers of $\log x$ and $\log(1-x)$.

6.4 A dyadic half-interval identity

The lower-half differential equation $F'(x) = 2F(2x)$ for $0 \leq x \leq 1/2$ leads to exact quantile truncations.

Proposition 6.2. *For every complex p , with removable values interpreted by continuation,*

$$\int_0^{1/2} Q(v)^p \, dv = 2^{-p} \frac{1 - \Re(p+1)}{p+1}, \quad (90)$$

$$\int_0^{1/2} x^p F(x) \, dx = \frac{2^{-p-1}}{p+1} \left[\frac{1}{2} - \frac{1 - \Re(p+2)}{p+2} \right]. \quad (91)$$

In particular,

$$\int_0^{1/2} F(x) \, dx = \frac{5}{72}. \quad (92)$$

Proof. Change variables $v = F(x)$ in the first integral and use the lower-half equation:

$$\int_0^{1/2} Q(v)^p \, dv = \int_0^{1/2} x^p F'(x) \, dx = 2^{-p} \int_0^1 y^p F(y) \, dy.$$

This is [equation \(90\)](#). Substitute it into the incomplete formula [equation \(30\)](#) at $x = 1/2$ to obtain [equation \(91\)](#). $\square$

7 Fractional and integer antiderivatives

Repeated integration has a particularly transparent probabilistic meaning. The same calculus extends without change from integer order to the Riemann–Liouville operator. For $\Re \alpha > 0$, write

$$(I_0^\alpha f)(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t) \, dt. \quad (93)$$

Integer versus fractional proof status

For $n \in \mathbb{N}$, the operator I_0^n here is the formal normalized Volterra operator J_0^n . In particular, Lean proves

$$I_0^n F(x) = 2^{\binom{n}{2}} F\left(\frac{x}{2^n}\right) \quad (x \leq 1),$$

including $n = 0$, and proves the analogous global identity for the signed extension. It also proves every natural-monomial weight as a finite dyadic sum. The probabilistic truncated-moment representation below, its inverse companion, and every noninteger α remain outside that formal result.

7.1 Fractional inverse-pair duality

Theorem 7.1 (fractional power duality). *Let $\Re \alpha > 0$. Initially for $\Re q > 0$,*

$$(I_0^\alpha F^q)(x) = \frac{q}{\Gamma(\alpha + 1)} \int_0^{F(x)} (x - Q(v))^\alpha v^{q-1} dv. \quad (94)$$

Initially for $\Re p > 0$,

$$(I_0^\alpha Q^p)(v) = \frac{p}{\Gamma(\alpha + 1)} \int_0^{Q(v)} (v - F(x))^\alpha x^{p-1} dx. \quad (95)$$

Both formulas continue in p or q wherever the endpoint terms admit continuation.

Proof. Since $d(x - t)^\alpha = -\alpha(x - t)^{\alpha-1} dt$, integration by parts gives

$$I_0^\alpha F^q(x) = \frac{1}{\Gamma(\alpha + 1)} \int_0^x (x - t)^\alpha d(F(t)^q).$$

The boundary terms vanish when $\Re q > 0$. Using $d(F^q) = qF^{q-1} dF$ and substituting $v = F(t)$ proves [equation \(94\)](#). The proof of [equation \(95\)](#) is identical after interchanging the inverse pair. $\square$

The case $q = 1$ has a direct truncated-moment interpretation.

Corollary 7.2. *For $\Re \alpha > 0$,*

$$(I_0^\alpha F)(x) = \frac{\mathbb{E}[(x - Y)_+^\alpha]}{\Gamma(\alpha + 1)}. \quad (96)$$

In particular,

$$(I_0^\alpha F)(1) = \frac{\mathfrak{M}(\alpha)}{\Gamma(\alpha + 1)}. \quad (97)$$

Proof. In [equation \(94\)](#), the integral over v is the expectation of $(x - Q(V))_+^\alpha$ for a uniform V . Since $Q(V) \stackrel{d}{=} Y$, this is [equation \(96\)](#). At $x = 1$, use $1 - Y \stackrel{d}{=} Y$. $\square$

Formula [equation \(97\)](#) says that the entire complex moment function is exactly the endpoint value of the fractional antiderivative, multiplied by $\Gamma(\alpha + 1)$. Thus the Newton series in [equation \(42\)](#) is at the same time a global series for fractional repeated integration.

7.2 Repeated primitives of Rvachev's up function

For $\Re \alpha > 0$, define the left-anchored fractional integral

$$(I_{-1}^\alpha \text{up})(x) = \frac{1}{\Gamma(\alpha)} \int_{-1}^x (x-t)^{\alpha-1} \text{up}(t) dt. \quad (98)$$

Because up is the density of Z ,

$$(I_{-1}^\alpha \text{up})(x) = \frac{\mathbb{E}[(x-Z)_+^{\alpha-1}]}{\Gamma(\alpha)}. \quad (99)$$

At the right endpoint,

$$(I_{-1}^\alpha \text{up})(1) = \frac{2^{\alpha-1} \mathfrak{M}(\alpha-1)}{\Gamma(\alpha)}. \quad (100)$$

For an integer $n \geq 1$, [equation \(99\)](#) becomes

$$(I_{-1}^n \text{up})(x) = \frac{1}{(n-1)!} \mathbb{E}[(x-Z)_+^{n-1}]. \quad (101)$$

7.3 Appell polynomial tails

For $x \geq 1$, the positive-part signs in [equations \(96\)](#) and [\(101\)](#) disappear. Hence

$$(I_0^n F)(x) = \frac{1}{n!} \mathbb{E}(x-Y)^n, \quad x \geq 1, \quad (102)$$

$$(I_{-1}^n \text{up})(x) = \frac{1}{(n-1)!} \mathbb{E}(x-Z)^{n-1}, \quad x \geq 1. \quad (103)$$

The polynomial sequences

$$P_n(x) = \mathbb{E}(x-Y)^n, \quad R_n(x) = \mathbb{E}(x-Z)^n \quad (104)$$

are Appell sequences:

$$P'_n(x) = nP_{n-1}(x), \quad R'_n(x) = nR_{n-1}(x). \quad (105)$$

Their exponential generating functions are

$$\sum_{n \geq 0} P_n(x) \frac{t^n}{n!} = e^{tx} P(-t), \quad (106)$$

$$\sum_{n \geq 0} R_n(x) \frac{t^n}{n!} = e^{tx} \mathcal{U}(-t). \quad (107)$$

These polynomial tails and their inverse-moment projection role are part of the existing Rvachev–Appell research in the repository; here they arise as the exterior part of the fractional-truncated-moment formula.

7.4 Ordinary n -fold antiderivatives as incomplete moments

For an integer $n \geq 1$, the anchored n -fold antiderivative of F is

$$I_0^n F(x) = \frac{1}{n!} \int_0^{F(x)} (x-Q(v))^n dv. \quad (108)$$

The same expression can be expanded finitely:

$$I_0^n F(x) = \frac{1}{n!} \sum_{j=0}^n (-1)^j \binom{n}{j} x^{n-j} \int_0^{F(x)} Q(v)^j dv. \quad (109)$$

Thus every repeated primitive is determined by incomplete quantile moments. At a dyadic x , these can in turn be reduced by the exact dyadic machinery already developed in the repository.

For inverse powers, [equation \(95\)](#) gives the complementary expression

$$I_0^n (Q^p)(v) = \frac{p}{n!} \int_0^{Q(v)} x^{p-1} (v - F(x))^n dx. \quad (110)$$

The special case $p = 1$ is a fractional extension of the area identity:

$$I_0^n Q(v) = \frac{1}{n!} \int_0^{Q(v)} (v - F(x))^n dx. \quad (111)$$

For $n = 1$, this is

$$\int_0^v Q(w) dw = \int_0^{Q(v)} (v - F(x)) dx, \quad (112)$$

the usual area formula in a form adapted to the Fabius quantile.

8 Derivatives of arbitrary order

8.1 Derivatives of the Fabius function

On the lower dyadic branch, repeated differentiation of $F'(x) = 2F(2x)$ gives

$$F^{(n)}(x) = 2^{n(n+1)/2} F(2^n x), \quad 0 < x < 2^{-n}. \quad (113)$$

The full interval is obtained by the known dyadic/Thue–Morse reflection formulas. We do not duplicate those formulas here; the role of [equation \(113\)](#) is to show why every inverse derivative becomes extremely singular near the endpoints.

8.2 Inverse derivatives and Bell polynomials

Let $f_j = F^{(j)}(x)$. Define rational differential expressions recursively by

$$\mathcal{R}_1(x) = \frac{1}{F'(x)}, \quad \mathcal{R}_{n+1}(x) = \frac{1}{F'(x)} \frac{d}{dx} \mathcal{R}_n(x). \quad (114)$$

Proposition 8.1 (all inverse derivatives). *For $0 < v < 1$,*

$$Q^{(n)}(v) = \mathcal{R}_n(Q(v)). \quad (115)$$

Equivalently, for $n \geq 2$,

$$Q^{(n)}(v) = -\frac{1}{F'(Q(v))} \sum_{k=2}^n F^{(k)}(Q(v)) B_{n,k}(Q'(v), \dots, Q^{(n-k+1)}(v)), \quad (116)$$

where $B_{n,k}$ is the partial exponential Bell polynomial.

Proof. The operator relation follows by applying $d/dv = (F'(x))^{-1} d/dx$ at $x = Q(v)$. The Bell recurrence follows by differentiating $F(Q(v)) = v$ n times and isolating the term containing $Q^{(n)}$. $\square$

The first expressions are

$$\begin{aligned} Q' &= \frac{1}{f_1}, \\ Q'' &= -\frac{f_2}{f_1^3}, \\ Q''' &= \frac{3f_2^2 - f_1 f_3}{f_1^5}, \\ Q'''' &= \frac{-15f_2^3 + 10f_1 f_2 f_3 - f_1^2 f_4}{f_1^7}, \end{aligned} \quad (117)$$

with every f_j evaluated at $x = Q(v)$. Combining these formulas with the Lambert– W inverse endpoint expansion from the repository gives asymptotics of any fixed derivative order.

8.3 Derivatives of integral transforms

Since $\mathfrak{M}$ is entire,

$$\mathfrak{M}^{(n)}(s) = \int_0^1 Q(v)^s (\log Q(v))^n dv. \quad (118)$$

The bilateral up transform satisfies, away from its complex zeros,

$$\mathcal{U}^{(n)}(t) = \mathcal{U}(t) B_n(L_1(t), \dots, L_n(t)), \quad (119)$$

where B_n is the complete exponential Bell polynomial and

$$L_j(t) = \frac{d^j}{dt^j} \log \mathcal{U}(t) = \sum_{k=1}^{\infty} 2^{-kj} \ell^{(j)}(t/2^k), \quad \ell(z) = \log \frac{\sinh z}{z}. \quad (120)$$

At $t = 0$, these are precisely the cumulants in [equation \(17\)](#); the continuation of [equation \(119\)](#) through zeros is obtained by cancelling the apparent logarithmic singularities.

9 Laplace, Fourier, Mellin, and resolvent transforms

9.1 One-sided Laplace transforms

Integration by parts against the distribution measure dF gives, for all $t \in \mathbb{C}$ with removable values at $t = 0$,

$$\int_0^1 e^{tx} F(x) dx = \frac{e^t - P(t)}{t}, \quad (121)$$

$$\int_0^1 e^{tx} (1 - F(x)) dx = \frac{P(t) - 1}{t}. \quad (122)$$

Consequently

$$\int_0^1 x^n e^{tx} F(x) dx = \frac{d^n}{dt^n} \left(\frac{e^t - P(t)}{t} \right), \quad (123)$$

$$\int_0^1 x^n e^{tx} \text{up}(x) dx = \frac{d^n}{dt^n} \left(\frac{P(t) - 1}{t} \right). \quad (124)$$

Purely imaginary t yields the corresponding sine and cosine transforms. On the full line,

$$\int_{-1}^1 e^{tx} \text{up}(x) dx = \mathcal{U}(t), \quad \int_{-1}^1 e^{i\omega x} \text{up}(x) dx = \phi(\omega). \quad (125)$$

9.2 Stieltjes transforms and all resolvent powers

For $z \in \mathbb{C} \setminus [0, 1]$, define

$$G_Y(z) = \mathbb{E} \frac{1}{z - Y} = \int_0^1 \frac{dv}{z - Q(v)}. \quad (126)$$

For $|z| > 1$,

$$G_Y(z) = \sum_{n=0}^{\infty} \frac{m_n}{z^{n+1}}. \quad (127)$$

Similarly, for $z \notin [-1, 1]$,

$$G_Z(z) = \int_{-1}^1 \frac{\text{up}(x)}{z - x} dx = \sum_{n=0}^{\infty} \frac{\mu_n}{z^{n+1}}, \quad |z| > 1. \quad (128)$$

Applying [equation \(77\)](#) to a resolvent primitive gives all higher powers. For every integer $n \geq 1$,

$$\int_0^1 \frac{F(x)}{(z - x)^{n+1}} dx = \frac{1}{n} \left[\frac{1}{(z - 1)^n} - \mathbb{E} \frac{1}{(z - Y)^n} \right], \quad (129)$$

$$\int_0^1 \frac{1 - F(x)}{(z - x)^{n+1}} dx = \frac{1}{n} \left[\mathbb{E} \frac{1}{(z - Y)^n} - \frac{1}{z^n} \right]. \quad (130)$$

For $n = 1$,

$$\int_0^1 \frac{F(x)}{(z - x)^2} dx = \frac{1}{z - 1} - G_Y(z), \quad (131)$$

$$\int_0^1 \frac{1 - F(x)}{(z - x)^2} dx = G_Y(z) - \frac{1}{z}. \quad (132)$$

Using the logarithmic primitive, with a branch analytic off $[0, 1]$,

$$\int_0^1 \frac{F(x)}{z - x} dx = \mathbb{E} \log(z - Y) - \log(z - 1), \quad (133)$$

$$\int_0^1 \frac{1 - F(x)}{z - x} dx = \log z - \mathbb{E} \log(z - Y). \quad (134)$$

These formulas furnish analytic continuations and boundary-value formulas for Hilbert transforms by approaching the cut from above and below.

9.3 Entire-kernel calculus

If $h(x) = \sum_{n \geq 0} a_n x^n$ is entire and the series is dominated on the compact support, then

$$\int_{-1}^1 h(x) \text{up}(x) dx = \sum_{n=0}^{\infty} a_n \mu_n, \quad (135)$$

$$\int_0^1 h(x) dF(x) = \sum_{n=0}^{\infty} a_n m_n. \quad (136)$$

For instance,

$$\int_{-1}^1 J_0(tx) \text{up}(x) dx = \sum_{n=0}^{\infty} \frac{(-1)^n (t^2/4)^n}{(n!)^2} \mu_{2n}. \quad (137)$$

Together with [equation \(77\)](#), this gives convergent moment series for Bessel, Airy, hypergeometric, and polylogarithmic kernels without requiring a separate quadrature of the flat endpoint function.

10 Products, order statistics, and correlation integrals

Integrals containing products of F and up are generally not rational, but they have exact probabilistic meanings that lead to useful transform formulas and asymptotics.

10.1 Products of distribution and survival powers

Let $Y_1, \dots, Y_{m+n}$ be independent copies of Y , and define

$$A_m = \max(Y_1, \dots, Y_m), \quad B_n = \min(Y_{m+1}, \dots, Y_{m+n}). \quad (138)$$

The two blocks are independent and

$$\mathbb{P}(A_m \leq x < B_n) = F(x)^m (1 - F(x))^n. \quad (139)$$

Theorem 10.1 (weighted order-statistic gap). *For integers $m, n \geq 1$ and $\Re p > -1$,*

$$\int_0^1 x^p F(x)^m (1 - F(x))^n dx = \frac{1}{p+1} \mathbb{E}[(B_n^{p+1} - A_m^{p+1}) \mathbf{1}_{\{A_m < B_n\}}]. \quad (140)$$

In particular,

$$\int_0^1 F(x)^m (1 - F(x))^n dx = \mathbb{E}(B_n - A_m)_+. \quad (141)$$

Proof. Insert the indicator of the event in [equation \(139\)](#), integrate, and apply Fubini. For each realization, the allowed interval is $[A_m, B_n)$ if $A_m < B_n$, and empty otherwise. Its x^p -integral is the right side of [equation \(140\)](#). $\square$

Since $\text{up}(x) = 1 - F(x)$ on $[0, 1]$, this theorem covers all integer products $F^m \text{up}^n$ on the positive half of the support. Two useful special cases are

$$\int_0^1 F(x)^m dx = \mathbb{E}(1 - A_m) = \mathbb{E} \min(Y_1, \dots, Y_m), \quad (142)$$

and

$$\int_0^1 F(x)^2 dx = \frac{1}{2} - \int_0^1 F(x)(1 - F(x)) dx. \quad (143)$$

When the density factor is present, substitution gives a closed beta value for arbitrary complex exponents in their convergence domain:

$$\int_0^1 F(x)^a (1 - F(x))^b F'(x) dx = B(a+1, b+1). \quad (144)$$

Using $F'(x) = 2 \text{up}(2x-1)$,

$$2 \int_0^1 F(x)^a (1 - F(x))^b \text{up}(2x-1) dx = B(a+1, b+1). \quad (145)$$

10.2 Quantiles and ordinary order statistics

If $Y_{r:n}$ is the r -th order statistic of n independent Fabius variables, then $Y_{r:n} = Q(V_{r:n})$ in distribution, where $V_{r:n}$ has beta density. Consequently

$$\mathbb{E}Y_{r:n}^p = \frac{1}{B(r, n-r+1)} \int_0^1 v^{r-1} (1-v)^{n-r} Q(v)^p dv. \quad (146)$$

Symmetry gives

$$\mathbb{E}Y_{r:n}^p = \mathbb{E}(1 - Y_{n+1-r:n})^p, \quad \mathbb{E}Y_{r:n} + \mathbb{E}Y_{n+1-r:n} = 1. \quad (147)$$

Expanding the beta polynomial in [equation \(146\)](#) reduces every integer n, r to a finite combination of the mixed quantile integrals appearing in [equation \(23\)](#).

10.3 The Gini integral and the sinc product

For two independent copies Y, Y' , the standard distribution identity is

$$\int_0^1 F(x)(1 - F(x)) dx = \frac{1}{2} \mathbb{E}|Y - Y'|. \quad (148)$$

The difference is $(Z - Z')/2$, and its characteristic function is $\phi(t/2)^2$. The absolute-moment formula

$$\mathbb{E}|X| = \frac{2}{\pi} \int_0^\infty \frac{1 - \Re \varphi_X(t)}{t^2} dt \quad (149)$$

therefore yields an integral containing only the infinite sinc product.

Theorem 10.2 (Gini–sinc identity). *With ϕ as in [equation \(11\)](#),*

$$\boxed{\int_0^1 F(x)(1 - F(x)) dx = \frac{1}{2\pi} \int_0^\infty \frac{1 - \phi(t)^2}{t^2} dt.} \quad (150)$$

Numerically,

$$\int_0^1 F(x)(1 - F(x)) dx = 0.0955632320160066 \dots, \quad (151)$$

so

$$\int_0^1 F(x)^2 dx = 0.4044367679839934 \dots \quad (152)$$

This is a direct link between a nonlinear real-space integral of F and the square of the Rvachev infinite product. It does not appear to simplify to the rational moment constants.

10.4 Autocorrelation and L^2 energy

Define

$$\mathcal{C}(a) = \int_{\mathbb{R}} \text{up}(x) \text{up}(x + a) dx. \quad (153)$$

Since up is even, the Wiener–Khinchin formula gives

$$\mathcal{C}(a) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-ita} \phi(t)^2 dt. \quad (154)$$

At $a = 0$, Parseval’s identity gives

$$\int_{\mathbb{R}} \text{up}(x)^2 dx = \frac{1}{\pi} \int_0^\infty \phi(t)^2 dt = 0.808873535967971 \dots \quad (155)$$

More generally, derivatives of $\mathcal{C}$ at the origin are convergent spectral moments whenever the corresponding weighted sinc-product integral converges; the fast decay results developed elsewhere in the repository provide quantitative truncation bounds.

11 Additional integral-transform identities

11.1 A transform relation for inverse resolvents

The general identity [theorem 3.1](#) can be applied in the v -variable as well. Taking $A(x) = x$ and $B(v) = \log(z/(z - v))$ gives

$$\int_0^1 \frac{Q(v)}{z - v} dv + \int_0^1 \log \frac{z}{z - F(x)} dx = \log \frac{z}{z - 1}. \quad (156)$$

With $B(v) = (z - v)^{-1}$,

$$\int_0^1 \frac{Q(v)}{(z - v)^2} dv + \int_0^1 \frac{dx}{z - F(x)} = \frac{1}{z - 1}. \quad (157)$$

These are nonlinear resolvent dualities: a transform of the quantile is paired with a composition transform of the distribution function.

11.2 Frullani representations for logarithmic constants

For $0 < Y \leq 1$,

$$-\log Y = \int_0^\infty \frac{e^{-tY} - e^{-t}}{t} dt. \quad (158)$$

Averaging and using $P(-t)$ gives

$$\int_0^1 \frac{F(x)}{x} dx = \int_0^\infty \frac{P(-t) - e^{-t}}{t} dt, \quad (159)$$

where

$$P(-t) = \prod_{k=1}^\infty \frac{1 - e^{-t/2^k}}{t/2^k}. \quad (160)$$

For $a > 0$, the negative moment itself has the Mellin–Laplace representation

$$\mathfrak{M}(-a) = \frac{1}{\Gamma(a)} \int_0^\infty t^{a-1} P(-t) dt. \quad (161)$$

This is an independent numerical route to [equation \(44\)](#) and connects the negative-moment constant C_* to the dyadic product on the positive Laplace axis.

11.3 A smoothing difference equation for $\mathfrak{M}$

Let

$$N(s) = \mathbb{E}(1 + Y)^s = \sum_{k=0}^\infty \binom{s}{k} m_k. \quad (162)$$

The series is entire because of [lemma 4.3](#). Integrating the uniform variable in [equation \(6\)](#) with an arbitrary complex exponent gives the entire difference relation

$$(s + 1)2^s \mathfrak{M}(s) = N(s + 1) - \mathfrak{M}(s + 1), \quad (163)$$

with the value at $s = -1$ interpreted by continuity. At nonnegative integer s , coefficient comparison recovers [equation \(14\)](#); at negative s , it is equivalent to [equation \(55\)](#). Thus the positive rational moments and the superexponentially growing negative moments are two sides of one entire difference equation.

12 Numerical experiments and reproducibility

The accompanying script `experiments.py` computes all numerical values in this report without direct sampling of Y . It uses four independent structures:

1. exact rational arithmetic for low moments via [equation \(14\)](#);
2. a positive, extended-precision moment recurrence based on a normalized $\text{Binomial}(n, 1/2)$ probability vector;

3. the Newton series [equation \(42\)](#), centered series [equation \(57\)](#), and stable negative recurrence [equation \(55\)](#);
4. adaptive quadrature of the sinc products in [equations \(150\)](#) and [\(155\)](#), with zeros supplied as break points and explicit algebraic tail bounds from the first several sinc factors.

The script used 12000 positive moments. Representative values are

Quantity	Numerical value
$\mathfrak{M}(1/2)$	0.696306844628413801
$\mathfrak{M}(-1/2)$	1.48925460566552167
$\mathfrak{M}(-1)$	2.31488614400581905
$-\mathbb{E} \log Y$	7.58240000404408657e-01
C_*	0.750721980407870959
$\int_0^1 F(1-F)$	0.0955632320160066
$\int_{\mathbb{R}} \text{up}(x)^2 dx$	0.808873535967971

The fractional values provide useful checks that are not forced by rational arithmetic:

$$\begin{aligned}
 \mathfrak{M}(1/2) &= 0.696306844628413801 \dots, \\
 \mathfrak{M}(-1/2) &= 1.48925460566552167 \dots, \\
 \mathfrak{M}(3/2) &= 0.368547739989599266 \dots, \\
 \mathfrak{M}(5/2) &= 0.213401104479966542 \dots
 \end{aligned} \tag{164}$$

The following samples of the periodic phase profile in [equation \(62\)](#) are strikingly close:

θ	$C(\theta)$
0.10	0.75072198040787095903
0.25	0.75072198040787095922
0.50	0.75072198040787096020
0.75	0.75072198040787096019
1.00	0.75072198040787095923

The spread is at the level of 10^{-18} , close to the effective precision of the long-double moment input; it motivates but does not prove [conjecture 13.1](#). The JSON file shipped with the report records all displayed data, the truncation size, and the Fourier cutoff.

12.1 Numerical error notes

The moment recurrence is cancellation-free. Truncating the Newton series after K terms is controlled by the superpolynomial tail $m_k = O_N(k^{-N})$; comparing $K = 6000, 8000, 10000, 12000$ stabilizes the displayed fractional values. For C_* , the recurrence itself supplies an a posteriori bound: after index r , the remaining relative product error is at most a constant multiple of $\sum_{j \geq r} 2^{1-j} = O(2^{-r})$.

For the Gini integral with cutoff T ,

$$\int_T^\infty \frac{1 - \phi(t)^2}{t^2} dt = \frac{1}{T} - \int_T^\infty \frac{\phi(t)^2}{t^2} dt.$$

Bounding the first three squared sinc factors gives an omitted correction $O(T^{-7})$. For the L^2 integral, five factors give an $O(T^{-9})$ tail. The script uses $T = 500$, making these errors smaller than the digits printed.

13 Conjectures and frontier directions

13.1 Near-rigidity of the negative-moment phase

Conjecture 13.1 (phase rigidity). *The continuous one-periodic limiting profile $C(\theta)$ in [equation \(62\)](#) is constant:*

$$C(\theta) \equiv C_*. \quad (165)$$

A weaker and perhaps more realistic alternative is that C is nonconstant but its first Fourier harmonics are suppressed by the same spectral mechanism responsible for the extremely small periodic oscillations in the endpoint asymptotics studied elsewhere in the repository. The exact recurrence reduces this question to the Fourier analysis of the periodic product [equation \(62\)](#). A rigorous high-precision test should compute $A(\theta)$ independently by [equation \(161\)](#), rather than by the same moment data used in the recurrence.

13.2 Completely regular growth of the entire Mellin function

The maximum modulus is known exactly through [equation \(72\)](#), but directional cancellation remains open.

Conjecture 13.2 (indicator function). *The entire function $\mathfrak{M}$ has completely regular growth of order 2 with indicator*

$$h(\vartheta) = \frac{\log 2}{2} \max\{0, -\cos \vartheta\}^2. \quad (166)$$

The upper bound follows immediately from $|\mathfrak{M}(re^{i\vartheta})| \leq \mathfrak{M}(r \max(0, -\cos \vartheta))$. The difficult part is a matching lower bound away from the negative real axis, where oscillation in $e^{i\Im(s) \log Y}$ may create zeros and exceptional directions.

13.3 Arithmetic nature of the new constants

The quantities

$$C_*, \quad \mathfrak{M}(-r), \quad -\mathbb{E} \log Y, \quad \int_0^1 F(1-F), \quad \int_{\mathbb{R}} \text{up}^2 \quad (167)$$

are defined by positive rational series or infinite sinc products, but no reduction to standard constants is visible.

Conjecture 13.3. *The constants C_* , $\mathfrak{M}(-r)$ for every positive integer r , and $-\mathbb{E} \log Y$ are irrational; at least one of the stronger properties of transcendence or algebraic independence should hold among them.*

This is deliberately stated as a research target, not as numerical evidence for a specific transcendence theorem. Possible tools include Mahler-type functional equations for $P(t)$, arithmetic estimates for the rational Newton partial sums, and Padé approximants controlled by the moment recurrence.

13.4 Complete monotonicity of normalized errors

The sequence C_r is decreasing by [equation \(66\)](#). Numerics suggest much more.

Conjecture 13.4. *The error sequence $C_r - C_*$ is completely monotone:*

$$(-1)^k \Delta^k (C_r - C_*) \geq 0 \quad (r \geq 1, k \geq 0). \quad (168)$$

A proof might represent $C_r - C_*$ as a Hausdorff moment sequence derived from the positive tail product $\prod_{j \geq r} (1 - \rho(j))$.

13.5 Order-statistic endpoint regimes

When the sample size n grows, the minimum $Y_{1:n}$ probes precisely the Lambert– W endpoint regime. Since

$$\mathbb{P}(Y_{1:n} > x) = (1 - F(x))^n,$$

its moments are

$$\mathbb{E}Y_{1:n}^p = p \int_0^1 x^{p-1} (1 - F(x))^n dx. \quad (169)$$

A promising direction is a uniform saddle-point expansion in p and n , with the periodic endpoint phase transported into order-statistic oscillations. The corresponding maximum follows by symmetry. This would connect the present integral calculus directly to the repository’s inverse-Lambert and half-integer spectral programs.

13.6 General base and q -deformations

Replace the dyadic weights by b^{-k} , normalized so the support remains $[0, 1]$. The distributional smoothing, inverse-pair calculus, and fractional identities survive unchanged, while the product $P(t)$, Bernoulli cumulants, and negative-moment normalization acquire base b . The expected leading law is

$$\mathbb{E}Y_b^{-r} \asymp C_b \frac{b^{r(r+1)/2}}{\Gamma(r)}. \quad (170)$$

Determining C_b , its phase profile, and its relationship with $q = 1/b$ binomial coefficients is a natural extension.

13.7 Formalization roadmap

The most accessible Lean targets are:

1. extend the formal support-free quantile pushforward to the two-function identity [theorem 3.1](#);
2. the power identity [equation \(23\)](#) and its polynomial corollaries;
3. the positive and negative moment recurrences;
4. positivity, monotonicity, and convergence of C_r ;
5. the order-statistic gap identity;
6. finite truncations and certified tail bounds for the Newton and sinc-product series.

The entire-function and fractional-integral portions require more analytic infrastructure, but can be decomposed into dominated-convergence, holomorphic-parameter integration, and beta-integral lemmas.

14 Summary of principal formulas

For convenient reference, the central formulas are collected here.

Distribution and transforms

$$Y = \sum_{k \geq 1} 2^{-k} U_k, \quad Z = 2Y - 1,$$

$$P(t) = \prod_{k \geq 1} \frac{e^{t/2^k} - 1}{t/2^k}, \quad \phi(t) = \prod_{k \geq 1} \frac{\sin(t/2^k)}{t/2^k},$$

$$\int_0^1 e^{tx} F(x) dx = \frac{e^t - P(t)}{t}, \quad \int_0^1 e^{tx} (1 - F(x)) dx = \frac{P(t) - 1}{t}.$$

Inverse-pair integration

$$A(z)B(F(z)) - A(0)B(0) = \int_0^z A'B \circ F + \int_0^{F(z)} A \circ Q B',$$

$$p \int_0^1 x^{p-1} F(x)^q dx + q \int_0^1 v^{q-1} Q(v)^p dv = 1,$$

$$s \int_0^1 e^{sx+tF(x)} dx + t \int_0^1 e^{tv+sQ(v)} dv = e^{s+t} - 1.$$

Entire moment calculus

$$\mathfrak{M}(s) = \int_0^1 Q(v)^s dv,$$

$$\mathfrak{M}(s) = \sum_{k \geq 0} (-1)^k \binom{s}{k} m_k = 2^{-s} \sum_{j \geq 0} \binom{s}{2j} \mu_{2j},$$

$$\int_0^1 x^{s-1} F(x) dx = \frac{1 - \mathfrak{M}(s)}{s}, \quad \int_0^1 x^{s-1} \text{up}(x) dx = \frac{\mathfrak{M}(s)}{s}.$$

Negative moments and fractional primitives

$$A(s+1) = \frac{2^{s+1}}{s} (A(s) - D(s)),$$

$$\mathfrak{M}(-r) \sim C_* \frac{2^{r(r+1)/2}}{\Gamma(r)},$$

$$I_0^\alpha F^q(x) = \frac{q}{\Gamma(\alpha+1)} \int_0^{F(x)} (x - Q(v))^\alpha v^{q-1} dv,$$

$$I_0^\alpha Q^p(v) = \frac{p}{\Gamma(\alpha+1)} \int_0^{Q(v)} (v - F(x))^\alpha x^{p-1} dx.$$

15 Conclusion

The probability/quantile viewpoint converts the integral theory of the Fabius–Rvachev system into a coherent calculus rather than a list of isolated evaluations. The inverse-pair identity organizes indefinite weighted integrals; the entire moment function organizes fractional and negative powers; Riemann–Liouville operators organize repeated antiderivatives; and the infinite sinc product organizes correlation integrals. The negative-moment recurrence is especially useful: it extracts a new constant and an analytic periodic phase directly from the dyadic smoothing law, with exponentially fast normalization.

The next decisive steps are to determine whether the phase profile is exactly constant, to connect its Fourier modes with the spectral corrections already present in endpoint asymptotics, and to

formalize the inverse-pair, fractional-order, and moment recurrences. The reusable integer-order Volterra layer and its exact Fabius specialization are already formal; these remaining directions are sufficiently concrete to support both high-precision experiments and Lean proofs.

A Exact low-order data

The exact positive moments used by the script are

n	$m_n = \mathbb{E}Y^n$	$\mu_n = \mathbb{E}Z^n$ when shown
0	1	1
1	1/2	0
2	5/18	1/9
3	1/6	0
4	143/1350	19/675
5	19/270	0
6	1153/23814	583/59535
7	583/17010	0
8	1616353/65063250	132809/32531625
9	132809/7229250	0
10	134926369/9762090030	46840699/24405225075

The centered entries can be generated more economically from $\mu_{2j} = 2m_{2j+1}/(2j+1)$ than by expanding $(2Y-1)^{2j}$, which avoids severe cancellation at high order.

B Repository audit categories

The recursive review used the following categories.

Category	Role in the audit
Core expositions	Definitions, probability law, smoothness, dyadic derivatives, symmetry, Fourier product, moment arithmetic.
Glossary and Lean walk-through	Normalization checks and correspondence with formalized objects.
Canonical research-frontier volume	Primary non-duplication comparison for asymptotics, Lambert- W , Bell/Bernoulli, q -binomial, interpolation, and extrapolation themes.
Current drafts	Checks against dyadic self-sampling, inverse-function expansions, Rvachev–Appell systems, half-integer spectral effects, Fourier decay, and certification algorithms.
Papers	Historical and arithmetic results, exact dyadic values, Fourier coefficients, and endpoint asymptotics.
Frozen archive	Historical variants and absorbed integration material, checked through the archive map and canonical successors to prevent counting old snapshots as independent frontiers.

C Reproducibility instructions

From the directory containing this report, run

```
python experiments.py --moments 12000 --cutoff 500
```

The script writes `numerical_results.json` and `numerical_results.tex`. The latter is included directly in the PDF table. Reducing `--moments` to 2000 gives a quick check; values involving the phase profile should use at least 10000 moments. The code requires NumPy, SciPy, and mpmath.

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